

Coherence-Weighted Electrical Drive in Choice and Motor Realization

A Receiver-Relative SAN Hypothesis with Preregistered Perturbation and Rescue Tests

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July 2026

Abstract

A nervous system must turn distributed candidate states into physical action without requiring a second inner agent to choose on its behalf. This paper develops one testable Self Aware Networks (SAN) hypothesis for that transition. It defines **SAN electrical valence** as a receiver-relative, signed projection of distributed oscillatory drive after source amplitude, transmission geometry, delay, inhibition, and receiver phase have been declared. The term does not mean psychological valence, valence-electron chemistry, or bulk net-charge accumulation in neural tissue. A cross-fitted departure of the proposed drive from a context-conditioned tonic reference is treated as one electrical projection of a Phase-Wave Differential (PWD). The resulting model distinguishes amplitude budget, effective phase concentration, signed receiver alignment, and unexpected drive; it therefore does not assume that greater synchrony always produces greater downstream influence.

Bat vocalization provides a bounded preparation-action-return case. Pre-vocal beta and high-gamma patterns predicted call class in a three-bat fronto-striatal study, and a later three-bat study found frequency- and epoch-specific changes in directed phase-transfer entropy before and after calls. Both papers explicitly limit causal interpretation. Separate human phase-triggered stimulation, mouse inhibitory-circuit, amygdala ensemble, and laryngeal-motor studies show that individual links in the proposed chain can be measured or manipulated, but none establishes the complete SAN operator. The paper therefore specifies nested predictive models, grouped validation, amplitude- and power-matched phase perturbations, component-specific loss, and preregistered rescue. It also separates represented-alternative selection from interruption of a prepared action loop. The hypothesis fails if its receiver-relative variable adds no held-out value beyond standard rate, power, burst, anatomy, body-state, effective-connectivity, and capacity-matched recurrent models, or if selective perturbation and rescue do not change the predicted preparation-to-action chain. The paper adds a preregistered synthetic transfer-model test across thirty-two replicates, forty sessions per replicate, and ten calibrated or misspecified estimators. It identifies the conditions under which delay error, missing sign, missing geometry, amplitude-only, and resultant-only models retain, lose, or falsely appear to explain the receiver-relative signal. This is a pre-data identifiability test, not animal evidence.

motor preparation; vocalization; receiver-relative coding; perturbation; reafference

Keywords: neural choice; phase-wave differential; electrical valence; oscillatory coherence;

Scope and evidence boundary

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Reader route: earn the vocabulary from the operation

Consider three ordinary events: deciding whether to speak, changing a prepared reach, or stopping an action that has already begun to form. In each case many neural populations may be active, yet only some

combination of their activity becomes effective at the particular downstream circuits that realize the action. The unexplained problem is not whether neurons fire or brain rhythms exist. It is how distributed signals with different amplitudes, signs, delays, orientations, and histories become a measurable influence at a named receiver.

Established neuroscience already supplies much of the surrounding chain. Excitatory and inhibitory inputs are integrated in a state-dependent manner; oscillatory phase and excitability can condition effective interaction; population dynamics can predict upcoming behavior; descending motor pathways transform cortical activity into motor-neuron recruitment; and movement or vocalization returns new sensory and bodily evidence. What remains open is whether one explicitly receiver-centered variable adds explanatory and causal value beyond ordinary rate, power, burst, connectivity, anatomy, and recurrent-state descriptions.

The proposed operation can be understood before it is named:

1. choose the receiver, prediction time, and candidate action;
2. express each source in the receiver's frame using declared amplitude, sign, gain, delay, and geometry;
3. combine those transformed sources and project the result onto the receiver's current directional or phase reference;
4. subtract the drive expected for that context using a reference learned without the held-out group;
5. test whether the remaining difference predicts action beyond serious alternatives; and
6. perturb the relevant component, measure its loss, restore it selectively, and trace the effect through motor output and returned consequence.

This paper calls the signed receiver projection **SAN electrical valence**. It calls the context-conditioned departure of that projection one electrical component of a **Phase-Wave Differential (PWD)**. The names summarize an operation already specified in ordinary language; they do not replace it.

1. The problem

Choice is often described at the level of options, values, intentions, or policies. Motor control is often described at the level of population state, corticospinal or corticobulbar recruitment, motor-unit activity, and force. A mechanistic bridge must say how a candidate state becomes able to influence a particular receiver and must remain distinguishable from ordinary firing-rate, spectral-power, movement-preparation, and recurrent-dynamics accounts.

This paper proposes such a bridge but does not claim that it is already established. The candidate sequence is:

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candidate recruitment
-> receiver-relative effective drive
-> departure from a context-conditioned tonic reference
-> route access and commitment
-> transformed descending motor recruitment
-> contraction, relaxation, or antagonist recruitment
-> returned sensory and bodily consequence
-> later criteria update
```

The narrow claim is that a declared receiver-relative drive may add predictive and causal content at the candidate-to-action transition. The broad SAN architecture, the complete typed PWD calculus, and the multiscale theory of choice are treated in companion papers. This paper owns the electrical- drive estimator and the preparation-to-muscle-to-return test.

2. Terms that must remain distinct

Several variables are easily collapsed by metaphor. They are separated here.

- **Psychological valence** is learned or innate appetitive-versus-aversive value.
- **Membrane voltage** is an electrical potential difference across a cell membrane.
- **Current** is charge flow through a specified compartment, synapse, membrane, or conductor.
- **Extracellular field potential** is a population-level measurement shaped by geometry and the measurement reference; it is not a direct inventory of transmitted charge.
- **Oscillatory phase** locates a signal within a declared cycle and reference.
- **Spectral power** summarizes variance attributed to a frequency range under an estimator.
- **Phase concentration or coherence** summarizes stable relative timing under a declared measure.
- **SAN electrical valence** is the new receiver-relative signed-drive hypothesis defined below.
- **PWD** is the broader receiver-relative difference event. Electrical valence supplies only one projection of that typed event.

The shared word *valence* supplies a historical analogy, not evidence that these quantities are the same. Appetitive value may bias which assembly is recruited. Electrical valence asks whether the recruited activity can exert signed influence at a named receiver.

The proposed variable also has a different job from adjacent vocabulary. Phase coding describes how events relate to a cycle; coherence describes stability of relative timing; a traveling wave describes propagation across space; and effective-connectivity measures estimate directed statistical influence under their stated models. The present operator composes signed sources through an explicit transfer model, projects them at a named receiver, subtracts a prospective context-conditioned reference, and then submits that difference to a preparation-to-action loss and rescue test. That joined receiver-relative operation is why the new term is useful.

3. Source genealogy and date boundary

The genealogy is staged rather than backfilled with the present equations. An SAN note internally associated with 2017 compared fish electric-field sensing with bat echolocation, described neurons as oscillatory send-and-return systems, and contrasted tonic expectation with phasic change. Its current public Git history begins on June 8, 2022; the internal date is owner chronology, not immutable public proof of the mature operator [11].

One locator requires an explicit filename boundary. Commit 37da575ba33f85cef4239295be161066d8d39f4c added the internally dated source under the original path `a0154z_ctpr.txt`; `a0154z.md` is the later canonical working filename. The public evidence route therefore uses the commit plus the original filename rather than constructing a nonexistent commit path from the later alias.

In August 2022, SAN directly incorporated the then-new bat information-flow paper. The note quoted the authors' directed phase-transfer-entropy result and retained their statement that the relation to vocal production remained correlational. A separate SAN note, first committed August 10, 2022, explicitly compared psychological valence with physical interaction, proposed that relative electrical valence could track neural-array phase changes, and linked higher recruited activity to behavior. It also cited the July 2022 BLA preprint that later became Piantadosi et al. [5]. A September 2022 note then described the bat result as an example of top-down control with PWD [11].

Those records establish a public 2022 genealogy for the joined valence, phase, and action idea. They do not establish the equations in this paper, demonstrate biological validity, or convert the internally dated 2017 text

into a 2017 public publication. The complex receiver model, cross-fitted reference, and perturbation-rescue estimands below are 2026 operationalizations.

4. Empirical constraints, not proofs of the whole theory

4.1 Bat vocal preparation and return

Weineck, Garcia-Rosales, and Hechavarria found call-related fronto-striatal oscillatory structure in three bats [1]. In one caudate contrast, beta power was higher for echolocation and gamma power was higher for communication. In frontal auditory field recordings, high gamma could instead be prominent before echolocation, with effects depending on cortical depth and epoch. The authors showed that pre-vocal activity could predict vocal output, but they explicitly called the evidence correlative and stated that it did not demonstrate a causal role for oscillations in vocal control. Their data are publicly routed by DOI 10.12751/g-node.6a0d94.

Garcia-Rosales and colleagues detected 12,494 calls in three bats and selected 138 echolocation and 734 communication calls preceded by at least 500 ms of silence [2]. Directed phase-transfer entropy (dPTE) showed predominant frontal-to-auditory directionality in spontaneous and pre-vocal periods, with a frequency- and call-dependent post-vocal reversal. The authors again stated that the relationship remained correlational and did not rigorously establish LFP causation of call initiation or planning. dPTE is a model-based directionality measure, not an intervention. The paper's public data route is 10.12751/g-node.q6xwhi.

These small-animal datasets are useful for within-animal development and falsification. They do not support unqualified population generalization. Any reanalysis must group folds and permutations by animal and session, report within-animal results, handle call imbalance, and avoid post-vocal leakage in a pre-vocal prediction endpoint.

4.2 Bounded causal component evidence

Stefanou and colleagues used state- and phase-triggered dual-coil TMS in 16 healthy participants [3]. Short-interval interhemispheric inhibition depended on the phase relation and excitability state of the measured mu rhythm. This is causal evidence that phase and state can condition an effective interaction in that preparation. It is not evidence for SAN electrical valence or a universal phase law.

Kuan and colleagues linked behavior, two-photon imaging, electron microscopy, and modeling in one animal [4]. Their reconstruction contained 124 excitatory and 16 inhibitory neurons and supported an opponent inhibitory motif related to choice selectivity. It is a valuable circuit constraint, not a universal choice algorithm.

Piantadosi and colleagues found stable, optically separable BLA ensembles associated with sucrose and quinine, used holographic photostimulation to change consummatory behavior bidirectionally, and reported functional antagonism and mutual inhibition between opposing ensembles [5]. That study supports graded population recruitment and causal behavioral valence in the BLA. It does not make psychological valence identical to electrical drive.

Human stopping studies add a temporal constraint. Wadsley and colleagues found evidence for a nonselective pause during selective stopping, while also finding that their inhibition signatures did not explain the magnitude of stopping interference [6]. Hervault and Wessel separated a common early pause from a later effector-specific retuning phase across three experiments [7]. These results support an extended control sequence, not a central chooser.

4.3 The motor chain is transformed

Neural activity does not deliver one bulk charge packet from cortex to muscle. Viral tracing in macaques and marmosets identified disynaptic cortical influence on laryngeal motoneurons and species-dependent premotor contributions [8]. Human imaging identified dorsal and ventral laryngeal representations with different cortical characteristics while controlling for breathing [9]. In eight Mandarin-speaking participants, direct cortical recordings and stimulation linked local laryngeal-motor-cortex populations to rising and lowering pitch dynamics [10]. These results constrain a multi-stage route from candidate state to motor-neuron recruitment, neuromuscular transmission, calcium-dependent muscle force, biomechanics, sound, and auditory return.

5. An identifiable receiver-relative model

5.1 Complex effective drive

For candidate assembly (c), let source (i) have nonnegative analytic amplitude $A_i(t)$, phase $\phi_i(t)$, and explicit excitatory or inhibitory sign $\sigma_{ic} \in \{-1, +1\}$. Let $G_{ic}(t) = g_{ic}(t)e^{-i\psi_{ic}(t)}$ be a complex transfer factor whose magnitude $g_{ic} \geq 0$ includes declared coupling and receiver sensitivity, and whose angle includes propagation delay, orientation, measurement reference, and other calibrated phase shifts. Define

$$Z_c(t) = \sum_{i=1}^{N_c} \sigma_{ic} G_{ic}(t) A_i(t) e^{i\phi_i(t)}. \quad (1)$$

The sign can equivalently be represented as a phase shift of π , but retaining it explicitly prevents inhibition from disappearing inside an opaque fitted phase. If G_{ic} is not measured or constrained, Equation (1) is only a latent model and cannot be advertised as an identified biophysical current.

Here σ_{ic} is an effective sign for the named source-receiver relation and analysis window, not an immutable label attached to a neurotransmitter or cell class. Reversal potentials, shunting inhibition, receiver state, and pathway recruitment can change the effective influence; any design that permits such changes must model $\sigma_{ic}(t)$ or stratify the relevant state rather than silently treating sign as fixed.

Let $\Phi_c(t)$ be the phase or directional reference of the named receiver. The signed receiver projection is

$$D_c(t) = \text{Re} \left[e^{-i\Phi_c(t)} Z_c(t) \right]. \quad (2)$$

Three companion quantities keep different sources of apparent gain separate:

$$B_c(t) = \sum_{i=1}^{N_c} g_{ic}(t) A_i(t), \quad (3)$$

$$R_c(t) = \frac{|Z_c(t)|}{B_c(t) + \varepsilon}, \quad 0 \leq R_c(t) \leq 1, \quad (4)$$

$$Q_c(t) = \frac{D_c(t)}{B_c(t) + \varepsilon}, \quad -1 \leq Q_c(t) \leq 1. \quad (5)$$

B_c is an amplitude-and-coupling budget, R_c is an effective resultant after sign and geometry, and Q_c is signed alignment with the receiver. Coherent sources can cancel if their effective signs or orientations oppose.

Splayed sources can still exert structured influence. Therefore synchrony is neither necessary nor sufficient for positive drive.

The units are also explicit. D_c and B_c inherit the declared source-proxy units after the transfer scaling, R_c and Q_c are dimensionless, and $\text{PWD}_{e,c}$ has the same units as D_c . Raw drive values from LFP, current, voltage, spike-derived, or imaging proxies cannot be compared across preparations without a declared calibration map. The normalized quantities improve comparability but do not by themselves identify the underlying biophysical transfer.

5.2 Tonic reference and electrical PWD

The tonic reference must be conditional, prospective, and learned without outcome leakage. Let X_t contain only preregistered information available before the prediction time, such as task state, body state, area, depth, trial history, and stable recording covariates. In each training fold estimate

$$m_c(X_t) = \mathbb{E}[D_c(t) \mid X_t]. \quad (6)$$

Apply that fitted reference to held-out observations and define

$$\text{PWD}_{e,c}(t) = D_c(t) - \hat{m}_c^{(-k)}(X_t), \quad (7)$$

where $(-k)$ marks a model fitted without the held-out group. This cross-fitted residual is one electrical projection of PWD, not the full typed PWD record. A baseline fitted on all trials, after seeing the outcome, or with post-action variables would invalidate the primary test.

5.3 Incremental prediction

For a preregistered prediction horizon δ , let $Y_{c,t+\delta} = 1$ indicate the candidate’s declared commitment or motor consequence. The primary nested model is

$$\text{logit Pr}(Y_{c,t+\delta} = 1) = \alpha + \beta_E \text{PWD}_{e,c}(t) + \gamma^\top \mathbf{z}_t + u_a + u_s, \quad (8)$$

where $\mathbf{z}_t$ contains preregistered rate, power, burst, anatomy, movement-preparation, body-state, task, and history controls, while u_a and u_s represent animal / participant and session effects when the design supports them. Separate ablations add B_c , R_c , and Q_c to determine whether any gain comes from amplitude, concentration, signed receiver alignment, or unexpected drive. Collinear features are not interpreted from unstable individual coefficients.

The primary predictive endpoint is grouped held-out improvement in negative log loss:

$$\Delta\mathcal{L} = \mathcal{L}_{\text{base}} - \mathcal{L}_{\text{full}}, \quad (9)$$

reported with calibration, uncertainty, and a frozen superiority margin. Accuracy alone is insufficient. The full model must also be compared with capacity-matched recurrent and state-space baselines.

5.4 Causal loss and rescue

Prediction does not establish that phase-sensitive drive controls action. Let δ_ϕ be a preregistered phase or route perturbation, 0 an energy- and sensory-matched sham, and δ_r a component-specific rescue. For outcome Y_c , define

$$\text{ACE}_{\text{loss}} = \mathbb{E}[Y_c \mid \text{do}(\delta_\phi)] - \mathbb{E}[Y_c \mid \text{do}(0)], \quad (10)$$

$$\text{ACE}_{\text{rescue}} = \mathbb{E}[Y_c \mid \text{do}(\delta_r)] - \mathbb{E}[Y_c \mid \text{do}(\delta_\phi)]. \quad (11)$$

Support requires target engagement, predicted loss, and directionally appropriate rescue without a matched change in nonspecific arousal, pain, movement artifact, or mean delivered energy. A broad behavioral disruption followed by recovery is not a component-specific rescue.

Equations (10)-(11) are estimands, not identification guarantees. Their interpretation requires a declared intervention assignment, consistency between the intervention and the stated treatment, positivity over the eligible units, measured target engagement, and control of differential missingness. If randomization is impossible, the paper must state the additional causal assumptions and treat the estimate as weaker. Rescue is a separate randomized contrast where feasible; it cannot be inferred merely because behavior returns toward baseline.

5.5 Pre-data transfer identifiability and failure regions

The first executable test asks a narrower question than the bat reanalysis: if Equation (1) is the data-generating model, which transfer errors can a grouped held-out analysis detect, and which errors leave a deceptively useful proxy? Thirty-two deterministic replicates each contain forty sessions, 120 trials per session, and eight signed sources. Sessions 1-20 train the tonic and outcome models, Sessions 21-30 are validation, and Sessions 31-40 remain sealed test sessions.

The generative outcome depends on the true cross-fitted receiver-relative drive together with task, body state, and amplitude budget. The analysis compares ten features:

1. calibrated sign, gain, transfer phase, and receiver phase;
2. transfer-phase error of 15 degrees;
3. transfer-phase error of 30 degrees;
4. transfer-phase error of 60 degrees;
5. transfer-phase error of 90 degrees;
6. a sign-blind transfer;
7. a geometry-free transfer;
8. amplitude budget alone;
9. resultant magnitude alone; and
10. normalized receiver alignment.

The complete sealed-test summary is retained in the machine-readable result. The principal medians are reproduced here so the interpretive claims can be checked without rerunning the program:

Estimator	Median correlation with true electrical PWD	Median retained log-loss gain
Calibrated transfer	1.0000	1.0000
Transfer error, 15 degrees	0.9970	1.0000

Estimator	Median correlation with true electrical PWD	Median retained log-loss gain
Transfer error, 30 degrees	0.9874	0.9916
Transfer error, 60 degrees	0.9283	0.9008
Transfer error, 90 degrees	0.7265	0.4980
Sign-blind transfer	-0.4596	0.5437
Geometry-free transfer	-0.0251	0.3231
Amplitude-budget feature	-0.0153	approximately 0
Resultant-magnitude feature	0.0222	approximately 0
Normalized receiver alignment	0.9910	0.9844

Retained predictive gain and signed correlation answer different questions. In particular, a sign-blind feature can retain predictive information because the fitted outcome coefficient may change sign, even while its direct orientation relative to the true electrical PWD is unstable. That behavior is one reason the paper preserves both diagnostics.

The calibrated model has a median sealed-test log-loss improvement of 0.0773 over the nuisance baseline; its fifth percentile is 0.0640. Delay error degrades correlation with the true PWD monotonically. Fifteen- and thirty-degree errors retain nearly all median predictive gain in this generative regime, sixty degrees retains about 90%, and ninety degrees retains about 50%. Consequently, a positive predictor does not by itself identify a precisely calibrated transfer angle.

Removing effective sign retains about 54% of calibrated median gain, while removing both sign and geometry retains about 32%. The initial assertion required sign blindness to retain less than 50% and failed. That failed receipt is preserved. Draft 3 therefore makes the narrower supported claim: sign is materially informative in this simulation, but the sign-blind proxy remains partially predictive and cannot be treated as carrying zero information.

Amplitude budget and resultant magnitude add no reliable incremental prediction once task, body state, and amplitude budget are already controlled. Normalized receiver alignment retains about 98% of calibrated median gain but changes its scale. This makes Q_c a serious compact comparator, not an expendable ablation.

The nuisance baseline already contains the amplitude budget. The amplitude-budget row is therefore a deliberate duplicate-information negative control: it tests whether relabeling amplitude as the candidate feature can manufacture incremental gain. It is not an independent test of whether amplitude ever matters biologically.

These results define failure regions for interpretation:

- small transfer-phase error can be prediction-equivalent even when the fitted angle is wrong;
- large phase error weakens but need not eliminate predictive value;
- missing sign or geometry can leave a partial signal;
- amplitude or coherence magnitude alone can fail despite apparently orderly activity; and
- receiver-normalized alignment can approximate the full predictor in a bounded regime.

The simulation supports the manuscript’s separation of B_c , R_c , Q_c , and $PWD_{e,c}$. It does not establish that the true biological transfer factors are known, that the bat recordings contain this variable, or that electrical valence causes action. Those claims still require calibrated anatomy and geometry, grouped biological data, target engagement, selective perturbation, and matched rescue.

6. Two choice families

6.1 Represented alternatives

A represented-alternative task declares two or more options before commitment. Sensory identity, learned consequence, body state, and motor requirements should be varied independently where possible. Candidate assemblies are decoded prospectively. The test asks whether the receiver- relative variables improve prediction of partial motor preparation and final action beyond values, rates, power, known anatomy, and a capacity-matched recurrent model.

The design must predeclare the options, receiver, precommitment window, commitment criterion, outcome, nuisance variables, grouping unit, and exclusion rules. An option decoded only after the movement begins is not evidence for a prospective choice mechanism.

6.2 Interruption of a prepared loop

Stopping or revising an already prepared action is a different task. A controlled stop / change paradigm should measure the initial trajectory, stop or change signal, frontocentral beta bursts, corticospinal excitability where appropriate, partial EMG, final action, replacement action, and later transfer. Wadsley et al. and Hervault and Wessel justify a pause-then-retune comparator [6,7].

Subjective awareness can be collected as a timing or access variable, but it is not the primary causal manipulation. Meditation, yoga, and clinical habit change are not required to test the operator and should not be used as the first-line evidence for it. If later studied, they require active controls for expectancy, instructor contact, breathing, movement, arousal, and practice.

7. Preregistered bat reanalysis

The bat analyses should be labeled exploratory unless preprocessing and endpoints are frozen before the held-out data are opened. A defensible development protocol is:

1. use only pre-vocal samples for call-class prediction;
2. group all inner and outer splits by animal and session;
3. report within-animal estimates because only three animals were studied;
4. stratify or weight the 138 echolocation and 734 communication calls without duplicating held-out trials;
5. estimate or remove aperiodic background and waveform-shape leakage before band comparisons;
6. include movement, call-acoustic, area, depth, state, and history covariates where observable;
7. compare rate-only, power / burst, phase-only, coherence, standard effective-connectivity, geometry-free, receiver-relative, state-space, and capacity-matched recurrent models;
8. use animal / session-blocked permutations and report calibration as well as discrimination; and
9. reserve a final sealed group only if enough independent groups remain to support it.

The public data routes in [1] and [2] make this pipeline auditable. They do not make a three-animal dataset large enough for an unqualified population claim.

7.1 Prospective biological protocol and dataset separation

The two bat papers support different observational tests and must not be pooled as if they measured one common receiver. The final package therefore includes a prospective protocol with two analysis families:

Family	Dataset and receiver relation	Primary predeclared question	Primary endpoint	Boundary
A	2020 fronto-striatal recordings	Does receiver-relative drive improve pre-vocal call-class prediction?	Animal / session-grouped held-out negative log loss, with calibration	Within-animal development evidence; no population claim from three bats
B	2022 fronto-auditory recordings	Does receiver-relative structure distinguish the pre-vocal top-down period from the post-call return period?	Grouped held-out epoch / direction prediction and prespecified dPTE-compatible comparators	Observational directionality; not causal call initiation

For both families the eligible calls, pre-vocal window, receiver definition, transfer-calibration procedure, exclusions, missingness rules, nuisance variables, model capacity, class weighting, random seeds, group assignments, and superiority rule must be frozen before any sealed group is opened. All preprocessing and residualization are learned inside training folds. Within-animal effects are reported separately; any across-animal summary is descriptive. The two datasets may cross-inform hypotheses, but their trials, receivers, and inferential units remain separate.

The protocol is intentionally prospective. This manuscript has not downloaded, transformed, or analyzed the three-bat recordings. A future analysis must issue a new execution receipt that names the exact dataset versions, exclusions, code hash, opened groups, and adverse as well as favorable results.

8. Beta-gamma is a secondary discriminator

The bat papers do not establish a global beta-to-gamma stencil or a universal negative correlation. The trial-level statistic must be local to animal (a), session (s), region (r), depth (d), call class (k), and predeclared time window (w). After fitting and removing declared nuisance structure from beta and gamma power, define

$$r_{BG}^{(a,s,r,d,k,w)} = \text{corr}(\tilde{P}_\beta, \tilde{P}_\gamma). \quad (12)$$

Residualization must occur inside the training folds. The nuisance model includes aperiodic slope and offset, waveform shape, broadband transients, movement, call acoustics, recording depth, region, state, and history. A hierarchical model may then estimate condition-specific effects without pooling incompatible sites. Failure to find negative trialwise association does not by itself refute receiver-relative drive; it refutes the preregistered beta-gamma subclaim.

9. From drive to muscle and returned consequence

For vocalization, the intermediate variables must be measured rather than skipped. Let d_t represent descending neural recruitment, n_t motor-neuron state, j_t neuromuscular transmission, q_t calcium-dependent muscle activation, and μ_t the agonist, antagonist, respiratory, and mechanical configuration. A measurement model is

$$M_t = H(d_t, n_t, j_t, q_t, \mu_t), \quad (13)$$

with outcomes such as laryngeal EMG, airflow, a validated vocal-fold proxy, and acoustic output. The action then changes the sensory and bodily input. Echo delay or structure can be manipulated independently

of preparation within ethical bounds, and ordinary laboratory actions can use outcome reversals. The critical return test is whether a changed consequence alters later receiver drive, value, route accessibility, or selection in the predicted direction.

10. Null models and falsification

The minimum comparator set is:

1. firing rate and spike count;
2. band power and burst probability;
3. amplitude budget without phase;
4. phase or coherence without geometry and sign;
5. standard functional or effective connectivity;
6. anatomy plus movement and body state;
7. reinforcement-learning value and policy variables;
8. capacity-matched state-space and recurrent models;
9. phase-shuffled, receiver-phase-shuffled, sign-scrambled, history-shuffled, and outcome-shuffled controls; and
10. nonspecific perturbation and sham controls.

The electrical-valence hypothesis is weakened or rejected in the tested preparation if any of the following preregistered conditions hold:

- receiver-relative features fail to exceed the frozen held-out superiority margin;
- apparent gain disappears after grouped validation or leakage correction;
- an amplitude-only, power-only, or equal-capacity recurrent model performs equivalently;
- the fitted transfer factors are not identifiable or stable enough to support interpretation;
- the prediction does not occur before commitment or motor onset;
- selective perturbation fails target engagement or produces no predicted component loss;
- rescue is absent, nonspecific, or explained by energy, arousal, pain, or movement;
- the proposed variable predicts action but not the measured preparation-to-muscle chain;
- returned-consequence manipulation does not produce the predicted later update; or
- results depend on one unregistered region, window, band, or post hoc reference.

An adverse result in the companion PWD work is already relevant: in synthetic capacity-matched tests, typed PWD did not beat established dynamic models in the declared approximation benchmark, and a raw recurrent model won another comparison [12]. The present paper does not reinterpret those failures as support. Its electrical projection earns standing only under a newly frozen biological protocol and a separate held-out endpoint.

10.1 Evidence-state ladder

The paper separates what has been executed from what has only been designed:

Level	Requirement	Present state
Component evidence	Published studies measure or perturb individual links in the chain	Source-audited and bounded
Synthetic identifiability	The declared estimator and misspecifications are tested under a known generative model	Executed; one failed assertion preserved

Level	Requirement	Present state
Biological incremental prediction	The receiver-relative feature beats frozen ordinary and capacity-matched alternatives on grouped held-out data	Prospective and unexecuted
Component loss	A selective intervention engages the target and changes the predicted link	Prospective and unexecuted
Component rescue	A matched intervention restores the specific lost component	Prospective and unexecuted
Motor-chain and return closure	Intermediate motor variables and returned consequences change as predicted	Prospective and unexecuted

Accordingly, the defensible endpoint of this version is a source-audited methods and model preprint with a validated synthetic analysis and an explicit biological protocol. It is not a completed animal experiment.

11. What would count as support

The strongest feasible result is a convergent pattern, not one significant coefficient:

identified receiver and calibrated transfer model
+ prospective held-out improvement over declared alternatives
+ stable result across grouped sessions and subjects or animals
+ selective phase / route perturbation with target engagement
+ directionally appropriate, component-specific rescue
+ measured propagation into the relevant motor chain
+ predicted change after returned-consequence manipulation

Any missing rung narrows the conclusion. Prediction without perturbation supports a biomarker. Perturbation without rescue supports a causal contribution but not the proposed component. Behavior change without the intermediate motor measurements supports neither the full chain nor the claimed receiver-relative mechanism.

12. Relationship to companion SAN papers

- **Choice Without a Central Chooser** owns the multiscale criteria-to-effective-route-to- commitment-to- reafference architecture. This paper supplies one candidate physical variable for the route-to-commitment transition.
- **Phase-Wave Differential Calculus** owns the typed PWD record, Neural Tuning update, Taylor / Volterra comparisons, synthetic simulations, and adverse results. This paper uses only the cross-fitted electrical projection in Equation (7).
- **Formal SAN Systems Whitepaper** owns the concise theory-wide summary. Its present scalar electrical-valence equation should be replaced or explicitly labeled as a pedagogical special case of Equations (1)-(7) after the companion owner reviews this draft.

This boundary prevents three different manuscripts from presenting the same operator as separate novelties.

13. Scope and conclusion

This paper does not prove free will, consciousness, NAPOT, or a universal neural code. It does not claim that psychological valence is electric charge, that coherence is always beneficial, that beta globally suppresses gamma, or that a field potential directly contracts a muscle. It does not attribute causal vocal initiation to correlational bat data.

Its contribution is an operational bridge. Distributed sources are passed through an explicit transfer model, projected onto a named receiver, separated into amplitude, resultant, alignment, and context-conditioned difference, and tested against serious alternatives. The hypothesis becomes choice-relevant only if the receiver-relative difference prospectively predicts and selectively controls the preparation-to-action-to-return sequence.

The resulting standard is deliberately difficult. If the model cannot beat simpler explanations, survive grouped validation, and support loss plus rescue, the electrical-valence component should be reduced or rejected in that preparation. The SAN genealogy remains historically documentable, but genealogy is not substituted for biological evidence.

14. Data, code, source, and release statement

This version reports no new participant or animal data. It packages the deterministic synthetic simulation program, its complete result, and the preserved initial failed-assertion receipt. The two public bat-data DOI routes are supplied for a future preregistered reanalysis, but no such reanalysis is represented as completed here.

The public-safe package contains the manuscript and PDF, a math-free provider description, the prospective biological protocol, a source-and-locator ledger, the simulation code and results, the failed-assertion receipt, a rich Claim-Source-Genealogy Atlas, a normalized paper-local fan-in export, validators, and exact manifests. Retained primary PDFs are evidence copies used for the source audit and are not redistributed in the release candidate. Their DOI routes, page locators, byte sizes, and SHA-256 hashes remain in the source ledger.

The local final-preprint package closes the work that can be completed without collecting or reanalyzing biological data. Ethics approval is required for any future intervention involving animals or people, but it is not a prerequisite for publishing this synthetic methods / model preprint. repository release, author submission, and external review remain separate actions.

Appendix A. Exact public source and artifact locators

Record	Exact public or packaged locator	Evidentiary use
Weineck et al. 2020	doi:10.1371/journal.pbio.3000658 ; data DOI 10.12751/g-node.6a0d94; PDF pp. 6, 13-17	Bounded pre-vocal fronto-striatal prediction and explicit correlational limit
Garcia-Rosales et al. 2022	doi:10.1038/s41467-022-31230-6 ; data DOI 10.12751/g-node.q6xwhi; PDF pp. 2, 4-7, 12-13	Bounded dPTE directionality, call counts, return reversal, and explicit causal limit
Stefanou et al. 2018	doi:10.1523/JNEUROSCI.1470-18.2018 ; PDF pp. 1, 7-9	Phase- and state-conditioned causal interaction in one TMS preparation
Kuan et al. 2024	doi:10.1038/s41586-024-07088-7 ; PDF pp. 2-9, 15	One-animal opponent inhibitory circuit constraint
Piantadosi et al. 2024	doi:10.1016/j.neuron.2023.11.007 ; PDF pp. 1, 6-13	Causal BLA ensemble and psychological-valence evidence

Record	Exact public or packaged locator	Evidentiary use
Wadsley et al. 2023	doi:10.1093/cercor/bhad239 ; PDF p. 1	Nonselective stopping pause and stated explanatory limit
Hervault and Wessel 2025	doi:10.1523/JNEUROSCI.1537-24.2025 ; PDF pp. 1, 11-13	Pause-then-retune action revision
Cerkevich et al. 2022	doi:10.1073/pnas.2122345119 ; PDF pp. 1-4	Cortical-to-laryngeal anatomical constraint
Eichert et al. 2020	doi:10.1093/cercor/bhaa182 ; PDF pp. 1-2, 9	Human dorsal and ventral laryngeal representations
Lu et al. 2023	doi:10.1038/s41467-023-42175-9 ; PDF pp. 1, 3-7	Local laryngeal-motor-cortex population and stimulation evidence
SAN send-return ancestor	https://github.com/v5ma/selfawarenetworks/blob/37da575ba33f85cef4239295be161066d8d39f4c/a0154z%20ctpr.txt	June 8, 2022 public fixation under the original filename; internal 2017 chronology remains separate
SAN valence-phase stage	https://github.com/v5ma/selfawarenetworks/blob/75b194fb97d2da09fb8bdd4130be98b910a72adb/a0271z.md	August 10, 2022 public valence / phase / behavior formulation
SAN bat-flow stage	https://github.com/v5ma/selfawarenetworks/blob/a3ad84338cd6529b2448e4452fdb3d82c76163bd/a0111z.md	August 19, 2022 bat-flow integration and correlational caveat
SAN bat / PWD stage	https://github.com/v5ma/selfawarenetworks/blob/73102828e0ecb88fbfe4b8ca903014b8079da105/a0001z.md	September 14, 2022 bat / PWD / top-down interpretation
Synthetic model	Packaged model/transfer-identifiability-simulation.py and result JSON	Executed pre-data identifiability analysis
Failed assertion	Packaged validation/transfer-identifiability-initial-failed-assertion.json	Preserved adverse threshold result

Appendix B. Final-preprint claim boundary

The manuscript advances a new receiver-relative operational hypothesis and an executable test program. Its positive evidentiary result is limited to the declared synthetic generative model. Published biological studies establish measurable component links, not the full operator. The biological reanalysis, selective perturbation, rescue, motor-chain propagation, and returned- consequence tests remain open empirical work.

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Authorship and research history

The SAN source program and the historical hypotheses described here were developed by Micah Blumberg. Public Git records establish the specific 2022 source stages identified in Section 3; earlier internally dated material is labeled separately. AI assisted under the author's direction with source retrieval, mathematical formalization, error finding, drafting, and verification. AI is not represented as an author or as the historical origin of SAN, PWD, NAPOT, Neural Rendering, or the Choice program. This current manuscript is locally complete as a source-audited methods and model paper. Electrophysiology, motor-systems, decision-neuroscience, statistics, and methods review can improve later versions but are not represented as already obtained.